



Review

Quantum biology of the retina

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ABSTRACT

The emerging field of quantum biology has led to a greater understanding of biological processes at the microscopic level. There is recent evidence to suggest that non-trivial quantum features such as entanglement, tunnelling and coherence have evolved in living systems. These quantum features are particularly evident in supersensitive light-harvesting systems such as in photosynthesis and photoreceptors. A biomimetic strategy utilizing biological quantum phenomena might allow new advances in the field of quantum engineering, particularly in quantum information systems. In addition, a better understanding of quantum biological features may lead to novel medical diagnostic and therapeutic developments. In the present review, we discuss the role of quantum physics in biological systems with an emphasis on the retina.

Key words: light signal transduction, photon, photoreceptor, quantum theory, retina.

INTRODUCTION

Quantum mechanics (QM) provides a more accurate theory to describe the behaviour of matter and energy than the classical theories that preceded it. The divergence between classical and quantum behaviour is most easily seen at very small scales.

QM is based on four key ideas: (1) all entities within the universe (e.g. a single particle, a human or even a galaxy) can only be properly described

with the aid of a quantum ‘wave function’. For the simple situation of a single particle, the square of the wave function at any location in the universe can be interpreted as the likelihood of finding the particle at that location; (2) that we have a tool called the Schrödinger equation, which allows us to evolve these wave functions in time; (3) that many key physical quantities, including energy and angular momentum, come in discrete units known as quanta; (4) that at the instant of observation (or measurement), an entity yields a certain result (the wave function is said to ‘collapse’), and thus the probabilistic wave function is no longer needed to describe the system.

The consequences of this inherently probabilistic description of the universe are manifold, but they imply: (a) that matter and energy behave in either wave-like or particle-like fashions depending on the type of observation; (b) that some properties of an entity are not knowable simultaneously with arbitrary accuracy – most famously the position and momentum of a particle; (c) objects can be in more than one state (e.g. location or energy) simultaneously – termed quantum superposition – which is eliminated when the system is measured.

These key principles of QM can lead to exotic consequences. Quantum *superposition* is well illustrated by Schrödinger’s thought experiment involving a cat inside a closed box in which the cat’s health has been tied to a radioactive decay event. According to QM, the cat exists in a ‘quantum superposition of states’ until the wave function collapses. Schrödinger initially proposed this in an attempt

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Received 3 March 2014; accepted 10 June 2014.

Competing/conflicts of interest: No stated conflict of interest.

Funding sources: No stated funding sources.

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to discredit the idea of superposition, but was disturbed to realize that his own mathematics supported it.¹

A loss of quantum superposition (*decoherence*) occurs when a system interacts with its environment in a thermodynamically irreversible way.^{2,3} This is equivalent to loss of information from the system to the environment. Quantum *tunnelling* describes a phenomenon where a particle is able to pass from one state to another even though it is energetically forbidden to do so (a property that is routinely used in electronics). A classical analogy of this occurs in thermally activated systems in which thermal fluctuations can allow a classical object to overcome a potential barrier that is higher than its average energy.

Quantum *entanglement* refers to a phenomenon whereby observations of one member of a pair of particles appear to enforce an outcome on a measurement of its partner, regardless of the pair's separation in time or space (Einstein called this 'spooky action at a distance').⁴ This has been demonstrated experimentally with photons, electrons and atoms.² These ideas make QM conceptually challenging, leading Niels Bohr (one of the founders of QM) to state: 'Anyone who is not shocked by quantum theory has not understood it'.⁵

However, despite the philosophical challenges, quantum physics has been singularly successful in explaining numerous observations that previously challenged classical physics. Furthermore, no experimental test has yet shown any failures in the theory. We are now entering an exciting period in which quantum physics is being applied to biology. In a broad sense, quantum physics explains all of chemistry at the subatomic scale and by extrapolation of much of biological chemistry. However, the term *quantum biology* is usually restricted to the nexus of biology with the more exotic features of QM, including entanglement, tunnelling and superposition. Evolution of life on Earth has had around 3.5 billion years to harness any survival benefits that QM might afford; hence, a better understanding of biological quantum processes may provide insights adaptable to quantum engineering, particularly quantum computing; conversely, it is not too far fetched to imagine that a better understanding of quantum biological processes may lead to medical diagnostic and therapeutic advances yet unimagined.

In this review, we discuss the role of QM in biological systems with an emphasis on the retina. We begin by presenting a brief overview of quantum biology and some examples where quantum effects are associated with biological function. The process of phototransduction in the retina and the quantum properties during this process are described. Finally, we discuss the potential applications that arise from the quantum biology of the retina.

QUANTUM BIOLOGY

The development of more advanced microscopes in the 20th century led to considerable interest in providing physical and mathematical descriptions of biological systems at the microscopic level. As a result, scientists then started to explore the role of quantum physics in biology in an attempt to explain the complexities of biological systems that classical physics could not account for. Two famous works that contributed to the advent of quantum biology are Schrödinger's 1944 classic, '*What Is Life?*',⁶ and Alan Turing's theory on morphogenesis, published in 1952.⁷ In his book, Schrödinger proposed that heredity is too accurate and stable to be accounted for by classical physics or statistical mechanics. He then postulated that genes were single molecules with the structure of 'aperiodic crystals', which were later confirmed to be DNA molecules with a linear sequence of non-repeating units.⁸ Turing's theory on morphogenesis was viewed as a prototype model for pattern formation of animal coats and skin pigmentation.⁷

Demonstration of quantum phenomena often requires highly isolated systems at extreme conditions. There is limited evidence of quantum phenomena surviving within biological environments. However, specialized organic structural abilities may have evolved to protect parts of the biological system from interactions with the environment, allowing the survival of quantum superposition over short time scales.⁹ Cai *et al.* in 2008 also hypothesized that a form of quantum entanglement may occur through persistent recurring entanglement in oscillating molecules.¹⁰

BIOLOGICAL SYSTEMS WITH QUANTUM PHENOMENA

The role of quantum phenomena in the process of photosynthesis is perhaps the best characterized area of research in quantum biology. In photosynthesis, energy in the form of photons is absorbed by light-harvesting antennas to be transported to a reaction centre for conversion to chemical energy. Several parts of the photosynthesis process, including electronic excitation transfer, may require quantum physics-based explanation. The **Fenna–Matthews–Olson complex of green sulfur bacteria** has been shown to play a significant role in the photosynthetic process by **mediating the transfer of light energy to the reaction centre via electronic excitation dynamics**. The remarkable efficiency of this complex allows a near 100% transfer of every absorbed photon to the reaction centre. Studies have shown that the Fenna–Matthews–Olson complex utilizes quantum properties, including superposition and long-lasting

coherence under physiological conditions, to achieve this remarkable rate of energy transfer without loss of information.^{11–13} Quantum tunnelling has also been shown to be evident as a widespread process in photosynthesis.^{14–17} This phenomenal efficiency of information transfer has attracted interest from the quantum engineering and computing community.

The ability to determine direction by sensing magnetic fields, so-called magnetoreception, as observed in some birds and other animals, has been shown through behavioural studies to involve a chemical reaction occurring in the retina of certain birds in response to the geomagnetic field. Schulten *et al.* in 1978 proposed a quantum mechanical explanation (the so-called ‘radical-pair mechanism’) to explain the light-dependent properties of magnetoreception.¹⁸ The radical-pair model is based on the spin of electrons and nuclei that can be found in different configurations. Although the role of quantum effects in this radical-pair model is still uncertain, recent studies have suggested the possibility of quantum coherence and entanglement of spin configurations.¹⁹

The mainstream theory that describes the mechanism of olfaction (The Shape theory) contends that different molecular shapes stimulate corresponding receptors (the lock and key model).²⁰ Recent studies have suggested that this traditional model is insufficient to explain the sensitivity of our olfaction. Turin, in 1996,²¹ proposed a signalling mechanism based on quantum tunnelling, where **tunnelling from an electron donor to an acceptor mediated by the odorant molecule adds a further selectivity to the olfactory process.** Recent investigations have provided evidence that such a mechanism is physically viable.²²

The retina is a light-sensitive tissue that converts incident light into electrical signals, which are then transmitted along the visual pathway to higher centres. The ability of biological photoreceptors to receive light at the level of single photons, as inferred by earlier studies,^{23,24} is itself a quantum phenomenon. Before discussing the quantum features of the retina, it is worth revising the process of phototransduction, which occurs immediately after light reception in the retina.

PHOTOTRANSDUCTION

Phototransduction is a process by which light energy is converted to electrical and chemical signals in the photoreceptors (Fig. 1). The human photoreceptors are packed densely in the outermost layer of the neural retina. The long axis of the rods and cones is aligned along the radii of the eye to maximize light collection.²⁵

Each photoreceptor is separated into a distinct outer and inner segment. The biochemical processes

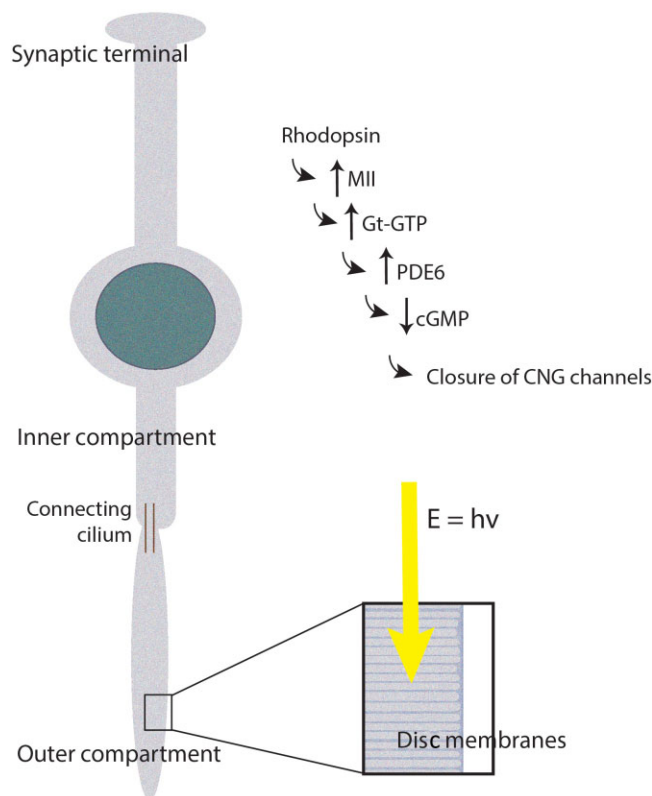


Figure 1. Schematics of rod cell response to light. cGMP, cyclic guanosine monophosphate; CNG, cyclic-nucleotide gated; Gt-GTP, G-protein guanosine triphosphate complex; MII, metarhodopsin II; PDE6, cGMP phosphodiesterase.

of phototransduction are limited to the outer segment. The outer segment has disc membranes that contain a different protein composition and a much higher concentration of photon receptor proteins compared with the cell’s plasma membrane. In rods, the disc membranes give an appearance of sealed vesicles, which are stacked up along the long axis of the rod’s outer segment and surrounded by cytoplasm and plasma membrane. The cones have a single membrane system where there are multiple invaginations of the plasma membrane in the outer segment, resembling disc membranes in rods.²⁵

The inner segment is the site of oxidative metabolism and active biosynthesis. **The outer and inner segments are connected by a very thin cilium, consisting of microtubules (MTs) surrounded by a thin layer of cytoplasm and plasma membrane.** Biochemical products are transported through this cilium from the inner segment to the outer segment for phototransduction. Photon particles captured in the outer segment will also pass through the inner segment and this bridging cilium. There is a constant passive communication of membrane potential between the inner and outer segments. When changes in membrane potential are transmitted from the outer segment through to the synaptic terminal,

neurotransmitter release is altered, resulting in signalling of the higher order neurons located at the inner layers of retina.²⁵

As rod photoreceptors have been much more widely studied than cones, the details of phototransduction described here in a general manner will be that which has been determined for rods. There are several important components that contribute to phototransduction, including 11-cis-retinal, rhodopsin, cyclic guanosine monophosphate (cGMP) gated sodium channel (also known as cyclic nucleotide-gated, CNG, channel), cGMP phosphodiesterase (PDE6) and G-protein (also known as transducin or Gt).²⁶

In the dark, when there is an absence of stimulus, outward movement of K⁺ ions is balanced against an inward Na⁺ current predominantly via non-gated K⁺ channels and CNG channels, respectively; these ionic gradients are also maintained via the action of the sodium-potassium adenosine triphosphatase (Na⁺/K⁺-ATPase) in the inner segment. Unusually for neuronal membranes, the open status of the CNG channel caused by elevated intracellular cGMP levels predominates and therefore the membrane potential of rods is maintained in a depolarized state at approximately -40 mV.^{25,27}

The CNG channel in the inner segment is a multi-subunit channel that has a greater probability of being open during the cellular inactive state, allowing Na⁺, K⁺ and Ca⁺ to pass. Binding of cGMP onto CNG channels is responsible for the increased probability of channel opening during this inactive state. The high level of intracellular cGMP present in photoreceptor outer segments during the inactive state is explained by there being reduced activities of cGMP phosphodiesterase. This is in turn due to the inactive Gt during the inactive state.²⁵

Rhodopsin is a photon receptor protein in the rod photoreceptors. It comprises 348 amino acids, in which mutations in several of the amino acid positions can lead to blinding conditions.²⁸ In the resting state, rhodopsin consists of the protein opsin and a covalently bound retinal molecule, which acts as a chromophore. In this state, the retinal molecule is in the 11-cis configuration.²⁵

Phototransduction commences with isomerization of the 11-cis retinal by incident photons. Isomerization of 11-cis retinal occurs at the bond between carbon 11 and 12, changing the molecular structure of 11-cis retinal into all-trans retinal. This isomerization of 11-cis retinal causes a conformational alteration to rhodopsin, which changes via a number of intermediates to metarhodopsin II.²⁹

Metarhodopsin II catalytically activates Gt, resulting in the release of bound guanosine diphosphate (GDP) from Gt. This is followed by Gt binding guanosine triphosphate (GTP) and forming a Gt-

GTP complex. The α subunit of this complex ($G\alpha$), upon dissociating from its bound $\beta\gamma$ subunits, activates cGMP phosphodiesterase, leading to hydrolysis and subsequent reduction in intracellular levels of cGMP. The outcome of this process is the closing of CNG channels, which no longer have enough free cGMP for their activation. Closure of the CNG channels results in alterations to transmembrane ionic gradients, which lead to hyperpolarization. This in turn reduces neurotransmitter release by the rods at the synaptic terminal, which is effectively interpreted by post-synaptic neurons as light absorption.^{25,29}

Upon photon absorption by rhodopsin, therefore, a cascade of events occur, which results in the hyperpolarization of the photoreceptor membranes, and it is this alteration in membrane potential, which ultimately initiates signalling to post-synaptic neurons.

QUANTUM FEATURES OF THE RETINA

Sensitivity to quantum properties of light

In 1942, Hecht *et al.*²⁴ proposed that two to seven photons are sufficient to be perceived by the human retina in the resting state. Since then, studies have suggested that the retina has the ability to perceive light at the level of a single photon.³⁰ However, in those experiments, conventional light sources such as light-emitting diodes were used, which possess intrinsic uncertainty in the number of emitted photons. Such fluctuations become fractionally larger for low intensity light pulses, which is exactly where the sensitivity of the rods are being assessed. Indeed, the single photon sensitivity of rods could only actually be inferred from statistical analyses of rod responses.^{23,30,31}

Displaying statistical properties

The distribution of fluctuations in the number of photons measured in a certain interval depends on the nature of the light source: coherent light from a laser has a different distribution to incoherent (thermal) light. Investigating the sensitivity of the retina with different types of light sources allows one to directly investigate whether the retinal response is preserving the quantum properties of the light source. Teich *et al.* in 1982 provided evidence that the human visual perception distinguishes between light sources with different photon statistics.³² The recent experiment by Sim *et al.* supported this notion and proposed that the variance of the responses of individual rods is the basis of this phenomenon.³¹

In Sim's experiment, retinas were taken from dark-adapted eyes of adult male frogs and single

rods were isolated into suction-electrode micropipettes. A fibre optic apparatus was used to deliver light pulses, and photoelectrical responses were amplified and analyzed.³¹

The experiment compared rod responses with excitation under coherent light and pseudothermal light sources. As hypothesized, the rods displayed a different pattern of photocurrent waveforms in response to photon fluctuations of various classical light sources. This study provided evidence for quantization of rod responses and the capability of rods to extract information about the quantum statistics of the light source.³¹

Conformational quantum superposition

Photoisomerization occurs at a remarkable rapid rate of <200 fs and proceeds at a very high efficiency or yield.³³ The process of photoisomerization involves transitioning of energy between different electronic states of the retinal molecule. Upon absorption of photons, the retinal molecule undergoes electronic transitioning from a resting ground state to a higher energy excited state. Further transfer of energy results in conformational change of rhodopsin and subsequently signal transduction. The remarkable speed and efficiency of photoisomerization is thought to depend on trivial quantum effects that shape the mechanical positioning of electronic states of the retinal molecule;^{33–36} however, conceivably, quantum effects, particularly coherence, could play a role in phototransduction.

Although quantum biology is usually associated with non-trivial quantum properties, as evidenced in the complexities of photosynthesis and avian magnetoreception, evidence of quantum superposition effects in the natural retina is still lacking. It is proposed that in a large protein, such as rhodopsin, random fluctuations among the molecular degrees of freedom are expected to cancel out any superposition effects.³⁷

Schlosshauer's work in 2006 postulated that upon receiving light, rhodopsin should still be able to achieve quantum superposition of different biochemically distinct states of the molecule. Coherent superposition of the cis and trans states of the retinal molecule may perhaps be achieved, albeit in a relatively fragile or transient state due to the influence of decoherence in a mesoscopic environment.³⁸

To investigate quantum coherent superposition in the human retina, Hilaire *et al.* proposed 'photon echo' experiments where a short laser pulse is sent to the retina followed by another pulse from the same source. If quantum coherence is occurring in the retina, persisting for duration that is longer than the time interval between the two laser pulses, a delayed 'photon echo' will be detected after the second pulse. This experiment questions the possibility of the

retinal molecule to exist in a coherent superposition of the cis and trans states. Currently, however, there exist no published results from these 'photon echo' experiments on the human retina.³⁹

Recent studies, however, have shown the possibility of quantum superposition in bacteriorhodopsin, which is the equivalent light-detecting protein system to rhodopsin, in Archaea. This effect was observed through the use of laser technology to achieve optical control of the retinal molecule in bacteriorhodopsin and change it from the all-trans to the 13-cis isomers under weak and high intensity fields. By applying pulse-shaped femtosecond laser excitation to the retinal in a native protein environment, Prokhorenko *et al.* achieved modulation of the isomerization yield by around 20%. This dependence on laser excitation indicates the relevance of quantum interference between vibrational states in the retinal moiety of bacteriorhodopsin.^{40,41}

didnot understand completely

Quantum tunnelling in mitochondria

The retina has prodigious energy demands. In the presence of oxygen, most of the cellular energy requirements are produced by oxidative phosphorylation in the mitochondria. Oxidative phosphorylation involves a series of redox reactions where electrons are transferred to oxygen, releasing energy that is used to generate adenosine triphosphate. The redox reactions take place in a series of protein complexes in the mitochondrial inner membrane known as the electron transport chain. Recently, it has been demonstrated that electrons are transferred in Complex I by quantum tunnelling.⁴²

This provides a remarkably efficient form of electron transfer. As mitochondrial impairment is a pathogenic feature of a number of age-related neurodegenerative diseases, including Parkinson's disease,⁴³ Alzheimer's disease,⁴⁴ Leber's hereditary optic neuropathy⁴⁵ and possibly glaucoma,^{46–48} it is interesting to speculate that a malfunction at the quantum level may be involved.

QUANTUM CONSCIOUSNESS AND THE RETINA

The nature of human consciousness remains a deep mystery and is a subject of ongoing research. A number of theories have been proposed to 'explain' consciousness. One such theory, which remains highly controversial, but deserves discussion, asserts that non-algorithmic phenomena exhibited by the human mind are explained by quantum computer processes. This so-called orchestrated 'objective reduction' theory was proposed by Hameroff and Penrose in the 1990s. They hypothesize that the MTs are the anatomical substrate for orchestrated objective reduction in the brain.^{49–51}

MTs, shaping the cytoskeleton of eukaryotic cells, are cylindrical polymers that self-assemble from tubulin proteins. MTs, such as centrioles and cilia, help to establish cell shape, direct growth and organize specific cell functions.⁴⁹ Centrioles are composed of nine sets of MT triplets, forming a shape of a cylinder. Albrecht-Buehler has shown that fibroblast cells communicate with each other via near infrared light and that their centrioles act as optical detectors.^{52,53} As centrioles play a major role in cell division and perform mirror-like activities with high precision, the possibility of quantum entanglement mediated by the quantum optical effects between centrioles has been suggested. Interestingly, cilia have a similar structure to centrioles as they are cylindrical in shape and consist of nine sets of MT doublets. Cilia have also been established as optical detectors in single cell organisms and primitive visual systems.⁵¹

The existence of quantum coherence in MTs is supported by a recently reported association between certain frequencies of MT vibrations under biological conditions and electroencephalographic readings as a function of consciousness.^{54,55} Quantum superposition could conceivably occur in MTs via quantum mechanical forces that act on individual tubulins. This would allow quantum communication between entangled tubulins in the same MT or neighbouring MTs. Hameroff and Penrose believe that the coherent superpositions of tubulins achieve decoherence or 'self-collapse' at the cerebral cortex and each of these events corresponds to a 'moment of conscious experience'.^{49–51,54,55}

MTs are widespread in the brain and also in the retina. Rod and cone cells contain a thin 'connecting cilium' that bridges the inner and outer segments of the cell. In fact, the light-sensitive portion of the photoreceptors is a modified primary cilium.⁵⁶ Photons that arrive at rhodopsin molecules in the outer segment would have first passed through these cilia. As they bear structural resemblance to centrioles, these cilia may act as waveguides for photons in the visible, ultraviolet and infrared spectra in human cells, while simultaneously extracting quantum information on polarization, angular and orbital momentum from photons.⁵¹ One of the predictions of the orchestrated objective reduction theory is that the MT-based cilia in rods and cones directly detect visual photons and connect with retinal glial cell MT via quantum tunnelling or entanglement.⁴⁹ The MTs then act as waveguides to transmit photons to the cerebral cortex in a quantum coherent manner, resulting in quantum visual perception and consciousness. To the best of our knowledge, the ability of MT-based cilia in the retinal rod and cone cells to detect photon quantum information remains untested.

APPLICATION OF QUANTUM BIOLOGY

The use of quantum physics is already evident in medical technologies such as the electron microscope and magnetic resonance imaging. Because quantum biology describes the involvement of quantum phenomena in living systems, there is significant interest in investigating and exploiting such features.

Of particular interest to quantum optics is the development of single photon emitters and detectors. The ability of the retina to be able to absorb and emit single photons and display different statistical properties in response to different photon statistics serves as a starting point for investigation and development of natural photoreceptors as photon detection or emission devices. Furthermore, the quantum processing of photoisomerization in different opsins occurs at remarkable rates and efficiency. Recent interest has been directed towards the octopus form of rhodopsin as a highly sensitive and efficient photon detector for applications in quantum communication and computation research.⁵⁷

With excellent photochemical efficiency and stability, the protein bacteriorhodopsin is by far the most widely researched natural photon receptor. It has been investigated for various applications, use in optical associative memories and processors.^{58,59} Furthermore, the recent work of Ahmadi *et al.* has suggested a potential role of bacteriorhodopsin as a radiation sensor in the medical field.⁶⁰

In the field of ophthalmology, bacteriorhodopsin remains under active research as a protein-based retinal prosthesis for patients suffering from retinal degenerative diseases. The proposed retinal implant mimics the light-absorbing capabilities of native human retina and generates a unidirectional proton gradient that is sufficient to stimulate the remaining functional retina.⁶¹

It is envisaged that pathological retinal photoreceptors may exhibit distinctive quantum signatures that could be exploited by novel diagnostic devices. Future research could investigate how the retina in different disease states may change its response.

CONCLUSION

Through experiments and theoretical studies performed over the past few decades, the understanding of quantum physics in the context of biology has allowed a better understanding of the processes occurring in the retina at atomic levels. Nevertheless, we have only scratched the surface of this exciting interdisciplinary research. A better understanding of the quantum properties of natural photoreceptors promises to yield rewards with both engineering and medical applications.

REFERENCES

1. Schrödinger E. Present situation in quantum mechanics. *Naturwissenschaften* 1935; **23**: 807–12, 23–28, 44–49.
2. Davies PC. Does quantum mechanics play a non-trivial role in life? *Biosystems* 2004; **78**: 69–79.
3. Miller WH. Perspective: quantum or classical coherence? *J Chem Phys* 2012; **136**: 210901.
4. Born M. *The Born-Einstein Letters: Correspondence between Albert Einstein and Max and Hedwig Born from 1916 to 1955*. New York: Walker, 1971.
5. Barad K. *Meeting the Universe Halfway: Quantum Physics and the Entanglement of Matter and Meaning*. Durham, NC: Duke University Press Books, 2007.
6. Schrodinger E. *What Is Life? The Physical Aspect of the Living Cell*. Cambridge: Cambridge University Press, 1944.
7. Turing AM. The chemical basis of morphogenesis. *Bull Math Biol* 1990; **52**: 153–97.
8. Watson JD, Crick FH. Molecular structure of nucleic acids. *Nature* 1953; **171**: 737–8.
9. Lidar DA, Chuang IL, Whaley KB. Decoherence-free subspaces for quantum computation. *Phys Rev Lett* 1998; **81**: 2594.
10. Cai J, Popescu S, Briegel HJ. Dynamic entanglement in oscillating molecules and potential biological implications. *Phys Rev E Stat Nonlin Soft Matter Phys* 2010; **82**: 021921.
11. Brixner T, Stenger J, Vaswani HM, Cho M, Blankenship RE, Fleming GR. Two-dimensional spectroscopy of electronic couplings in photosynthesis. *Nature* 2005; **434**: 625–8.
12. Engel GS, Calhoun TR, Read EL et al. Evidence for wavelike energy transfer through quantum coherence in photosynthetic systems. *Nature* 2007; **446**: 782–6.
13. Panitchayangkoon G, Hayes D, Fransted KA et al. Long-lived quantum coherence in photosynthetic complexes at physiological temperature. *PNAS* 2010; **107**: 12766–70.
14. Blankenship RE. *Tunneling Processes in Photosynthesis Research: Special Issue, Volume 2*: Kluwer Academic Publishers, 1989.
15. Hildner R, Brinks D, Nieder JB, Cogdell RJ, van Hulst NF. Quantum coherent energy transfer over varying pathways in single light-harvesting complexes. *Science* 2013; **340**: 1448–51.
16. Lambert N, Chen Y-N, Cheng Y-C, Li C-M, Chen G-Y, Nori F. Quantum biology. *Nat Phys* 2012; **9**: 10–8.
17. Ringsmuth AK, Milburn GJ, Stace TM. Multiscale photosynthetic and biomimetic excitation energy transfer. *Nat Phys* 2012; **8**: 562–7.
18. Schulten K, Swenberg CE, Weller A. A biomagnetic sensory mechanism based on magnetic field modulated coherent electron spin motion. *Z Phys Chem* 1978; **111**: 1–5.
19. Stoneham AM, Gauger EM, Porfyakis K, Benjamin SC, Lovett BW. A new type of radical-pair-based model for magnetoreception. *Biophys J* 2012; **102**: 961–8.
20. Buck L, Axel R. A novel multigene family may encode odorant receptors: a molecular basis for odor recognition. *Cell* 1991; **65**: 175–87.
21. Turin L. A spectroscopic mechanism for primary olfactory reception. *Chem Senses* 1996; **21**: 773–91.
22. Brookes JC, Hartoutsiou F, Horsfield A, Stoneham A. Could humans recognize odor by phonon assisted tunneling? *Phys Rev Lett* 2007; **98**: 038101.
23. Baylor DA, Lamb TD, Yau KW. Responses of retinal rods to single photons. *J Physiol* 1979; **288**: 613–34.
24. Hecht S, Shlaer S, Pirenne MH. Energy, quanta, and vision. *J Gen Physiol* 1942; **25**: 819–40.
25. Burns ME, Arshavsky VY. Beyond counting photons: trials and trends in vertebrate visual transduction. *Neuron* 2005; **48**: 387–401.
26. Arshavsky VY, Lamb TD, Pugh EN Jr. G proteins and phototransduction. *Annu Rev Physiol* 2002; **64**: 153–87.
27. Matte SL, Laue TM, Cote RH. Characterization of conformational changes and protein-protein interactions of rod photoreceptor phosphodiesterase (PDE6). *J Biol Chem* 2012; **287**: 20111–21.
28. Hernández-Rodríguez EW, Sánchez-García E, Crespo-Otero R, Montero-Alejo AL, Montero LA, Thiel W. Understanding rhodopsin mutations linked to the retinitis pigmentosa disease: a QM/MM and DFT/MRCI study. *J Phys Chem B* 2012; **116**: 1060–76.
29. Bartl FJ, Vogel R. Structural and functional properties of metarhodopsin III: recent spectroscopic studies on deactivation pathways of rhodopsin. *Phys Chem Chem Phys* 2007; **9**: 1648–58.
30. Rieke F, Baylor DA. Origin of reproducibility in the responses of retinal rods to single photons. *Biophys J* 1998; **75**: 1836–57.
31. Sim N, Cheng MF, Bessarab D, Jones CM, Krivitsky LA. Measurement of photon statistics with live photoreceptor cells. *Phys Rev Lett* 2012; **109**: 113601.
32. Teich MC, Prucnal PR, Vannucci G, Breton ME, McGill WJ. Multiplication noise in the human visual system at threshold. *Biol Cybern* 1982; **44**: 157–65.
33. Ben-Nun M, Molnar F, Lu H, Phillips JC, Martínez TJ, Schulten K. Quantum dynamics of the femtosecond photoisomerization of retinal in bacteriorhodopsin. *Faraday Discuss* 1998; **110**: 447–62.
34. Abe M, Ohtsuki Y, Fujimura Y, Domcke W. Optimal control of ultrafast cis-trans photoisomerization of retinal in rhodopsin via a conical intersection. *J Chem Phys* 2005; **123**: 144508.
35. Hayashi S, Tajkhorshid E, Schulten K. Photochemical reaction dynamics of the primary event of vision studied by means of a hybrid molecular simulation. *Biophys J* 2009; **96**: 403–16.
36. Polli D, Altoè P, Weingart O et al. Conical intersection dynamics of the primary photoisomerization event in vision. *Nature* 2010; **467**: 440–3.
37. Arndt M, Juffmann T, Vedral V. Quantum physics meets biology. *HFSP J* 2009; **3**: 386–400.
38. Schlosshauer M. Experimental motivation and empirical consistency in minimal no-collapse quantum mechanics. *Ann Phys (N Y)* 2006; **321**: 112–49.
39. Hilaire PS, Bierman D, Hameroff S. Quantum states in the retina. 2002.
40. Prokhorenko VI, Halpin A, Johnson PJ, Miller R, Brown LS. Coherent control of the isomerization of

- retinal in bacteriorhodopsin in the high intensity regime. *J Chem Phys* 2011; **134**: 085105.
41. Prokhorenko VI, Nagy AM, Waschuk SA, Brown LS, Birge RR, Miller RD. Coherent control of retinal isomerization in bacteriorhodopsin. *Science* 2006; **313**: 1257–61.
 42. Hayashi T, Stuchebrukhov AA. Electron tunneling in respiratory complex I. *Proc Natl Acad Sci U S A* 2010; **107**: 19157–62.
 43. Hauser DN, Hastings TG. Mitochondrial dysfunction and oxidative stress in Parkinson's disease and monogenic parkinsonism. *Neurobiol Dis* 2013; **51**: 35–42.
 44. Mancuso C, Scapagini G, Curro D *et al.* Mitochondrial dysfunction, free radical generation and cellular stress response in neurodegenerative disorders. *Front Biosci* 2007; **12**: 1107–23.
 45. Man PY, Turnbull DM, Chinnery PF. Leber hereditary optic neuropathy. *J Med Genet* 2002; **39**: 162–9.
 46. Chrysostomou V, Rezaei F, Trounce IA, Crowston JG. Oxidative stress and mitochondrial dysfunction in glaucoma. *Curr Opin Pharmacol* 2013; **13**: 12–5.
 47. Lee S, Sheek L, Crowston JG *et al.* Impaired complex-I-linked respiration and ATP synthesis in primary open-angle glaucoma patient lymphoblasts. *Invest Ophthalmol Vis Sci* 2012; **53**: 2431–7.
 48. Schober MS, Chidlow G, Wood JP, Casson RJ. Bioenergetic-based neuroprotection and glaucoma. *Clin Experiment Ophthalmol* 2008; **36**: 377–85.
 49. Hameroff S, Penrose R. Consciousness in the universe: a review of the 'Orch OR' theory. *Phys Life Rev* 2014; **11**: 39–78.
 50. Hameroff S, Penrose R. Orchestrated reduction of quantum coherence in brain microtubules: a model for consciousness. *Math Comput Simul* 1996; **40**: 453–80.
 51. Hameroff SR, Tuszynski JA, eds. Quantum states in proteins and protein assemblies: The essence of life? In: *Second International Symposium on Fluctuations and Noise*. Maspalomas, Gran Canaria Island, Spain: International Society for Optics and Photonics, 2004; 27–41.
 52. Albrecht-Buehler G. Cellular infrared detector appears to be contained in the centrosome. *Cell Motil Cytoskeleton* 1994; **27**: 262–71.
 53. Albrecht-Buehler G. Changes of cell behavior by near-infrared signals. *Cell Motil Cytoskeleton* 1995; **32**: 299–304.
 54. Sahu S, Ghosh S, Ghosh B *et al.* Atomic water channel controlling remarkable properties of a single brain microtubule: correlating single protein to its supramolecular assembly. *Biosens Bioelectron* 2013; **47**: 141–8.
 55. Sahu S, Ghosh S, Hirata K, Fujita D, Bandyopadhyay A. Multi-level memory-switching properties of a single brain microtubule. *Appl Phys Lett* 2013; **102**: 123701.
 56. Insinna EM. Ciliated cell electrodynamics: from cilia and flagella to ciliated sensory systems. In: Malhotra SK, Tuszynski JA, eds. *Advances in Structural Biology*, Vol. 5. Stamford, CT: JAI Press, 1998; 9–42.
 57. Sivozhelezov V, Nicolini C. Prospects for octopus rhodopsin utilization in optical and quantum computation. *Phys Part Nuclei Lett* 2007; **4**: 189–96.
 58. Greco JA, Wagner NL, Ranaghan MJ, Rajasekaran S, Birge RR. Biomolecular electronics and protein-based optical computing. In: Katz E, ed. *Biomolecular Information Processing: from Logic Systems to Smart Sensors and Actuators*. Weinheim, Germany: Wiley-VCH, 2012; 33–59.
 59. Wagner NL, Greco JA, Ranaghan MJ, Birge RR. Directed evolution of bacteriorhodopsin for applications in bioelectronics. *J R Soc Interface* 2013; **10**: 20130197, 1–15.
 60. Ahmadi M, Osei EK, Yeow JT. Bacteriorhodopsin for superficial X-ray sensing. *Sens Actuators B Chem* 2012; **166**: 177–83.
 61. Wagner NL, Greco JA, Birge RR. Visual restoration using microbial rhodopsins. In: Rehm BHA, ed. *Bionanotechnology: Biological Self-Assembly and its Applications*. Norfolk: Caister Academic Press, 2013; 205–40.